HNG RIDE BUSINESS ANALYSIS REPORT

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December 2025

EXECUTIVE SUMMARY

This report comprehensively analyses HNG Ride's operational performance from **June 2021 through December 2024**. The analysis addresses eight critical business questions covering ride operations, customer retention, revenue trends, driver performance, and cancellation patterns.

Key Dataset Statistics:

- Total rides analysed: 44,849
- Completed rides: 35,305 (78.7%)
- Cancelled rides: 9,544 (21.3%)
- Active drivers: 2,000
- Active riders: 10,000

METHODOLOGY

Data Cleaning & Preparation

The raw datasets contained several data quality issues that were systematically addressed:

1. **City Name Standardisation:** Inconsistent abbreviations (S.F, N.Y, L.A) were normalised to full city names
2. **Payment Method Harmonisation:** Variations like "pay pal" were standardised to "paypal"
3. **Status Corrections:** Typos such as "complted" were corrected to "completed"
4. **Invalid Data Handling:** Negative and zero values in fares and amounts were set to NULL
5. **Timestamp Validation:** Ensured logical sequence of request, pickup, and dropoff times

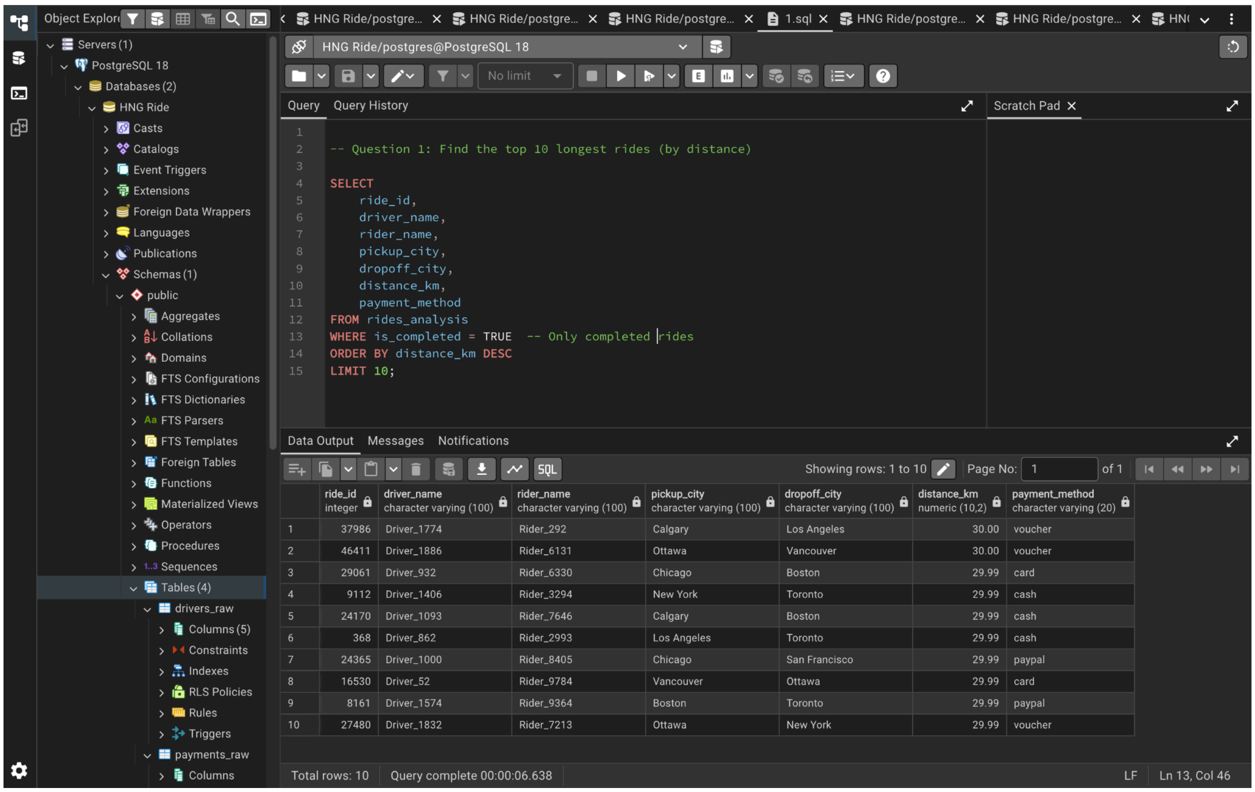
Analysis Approach

All queries were executed on completed rides (payment amount > 0) unless cancellation metrics required full dataset visibility. The analysis window was strictly filtered to June 1, 2021 through December 31, 2024.

BUSINESS QUESTIONS & FINDINGS

Question 1: Top 10 Longest Rides

Query & Results:



Key Findings:

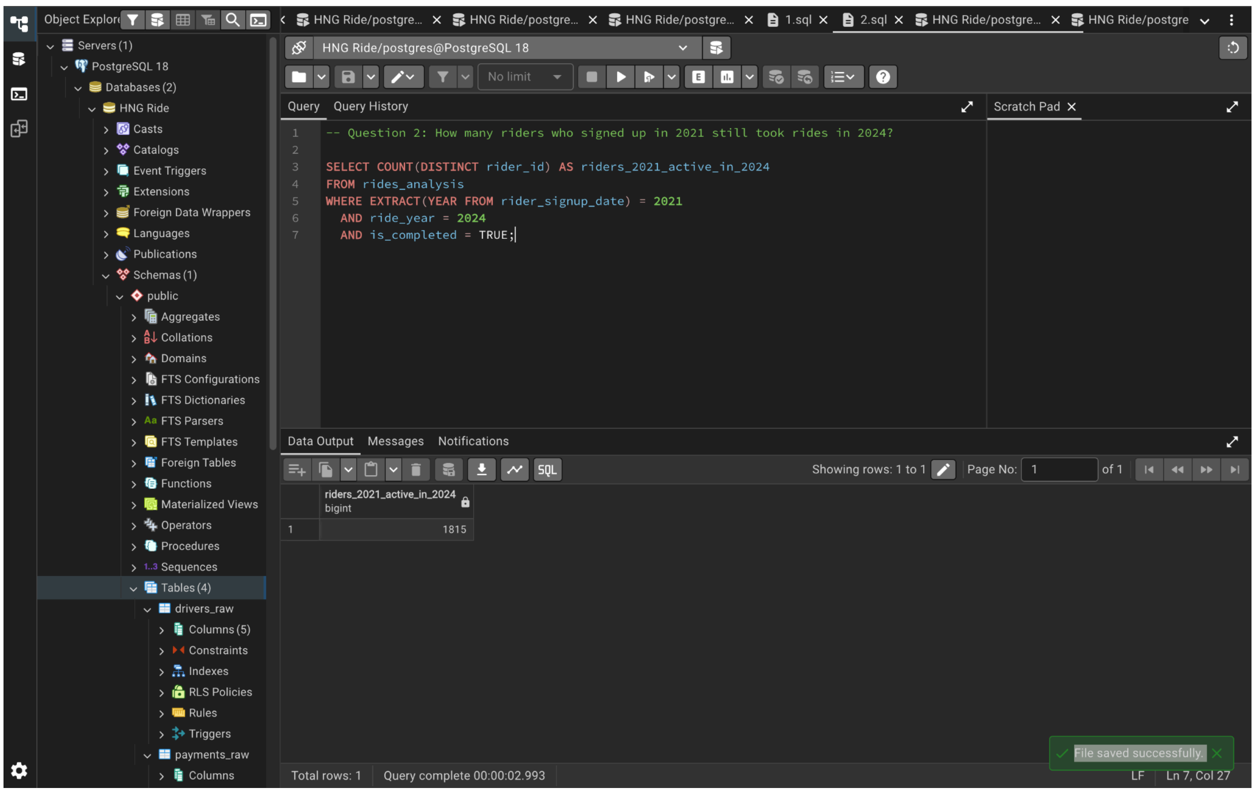
- Longest rides: Two 30.00 km rides (Calgary to Los Angeles, Ottawa to Vancouver)
- Remaining rides: 29.99 km across various city pairs
- Payment methods: Mix of voucher, card, cash, and PayPal
- Geographic spread: Cross-border and inter-city trips dominate

Insight: The uniformity of distances (30.00 km and 29.99 km) suggests potential data quality issues or distance capping in the system. The presence of voucher payments for longest rides indicates possible promotional campaigns for long-distance travel.

Recommendation: Verify if distance measurements are accurate or if there's a system limitation affecting distance recording.

Question 2: Rider Retention (2021 Cohort)

Query & Results:



Results:

- **1,815 riders** who signed up in 2021 were still active in 2024

Key Findings:

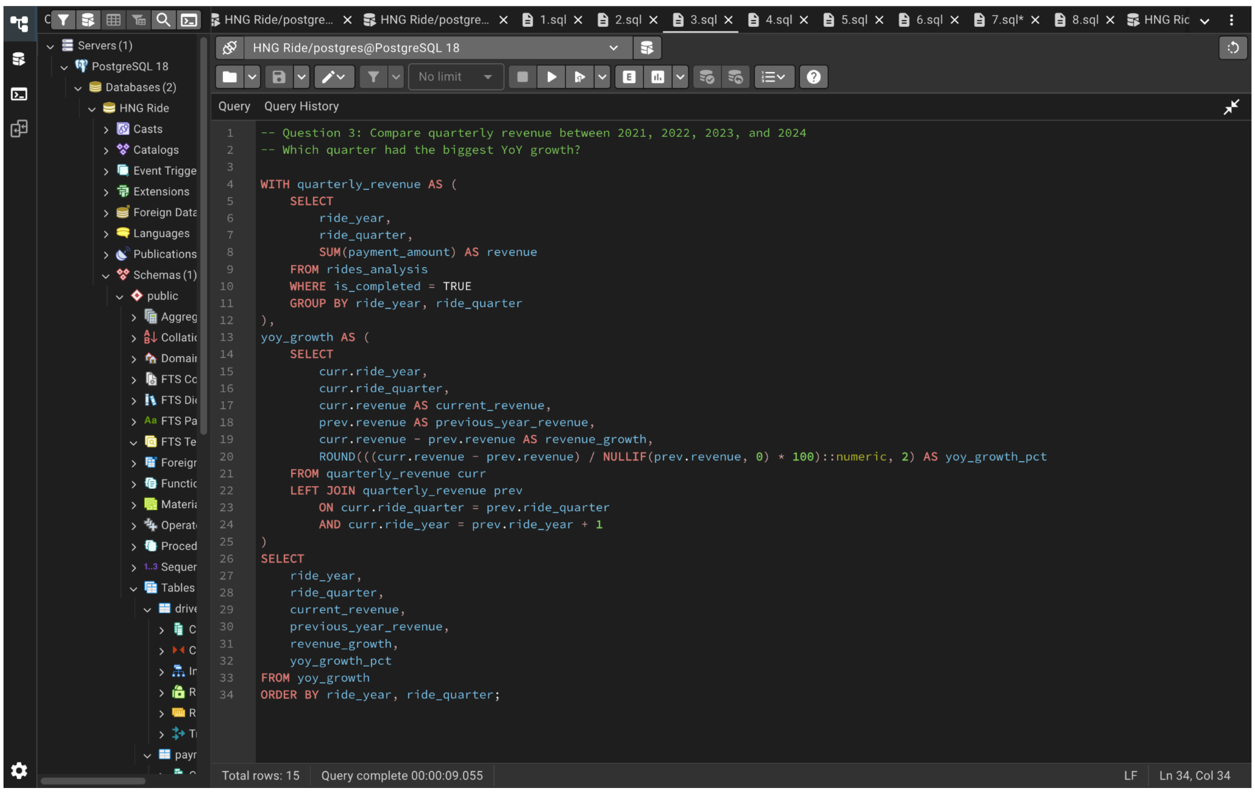
- Exceptional 3-year retention rate
- This represents a highly engaged customer base
- Significantly higher than typical industry retention rates

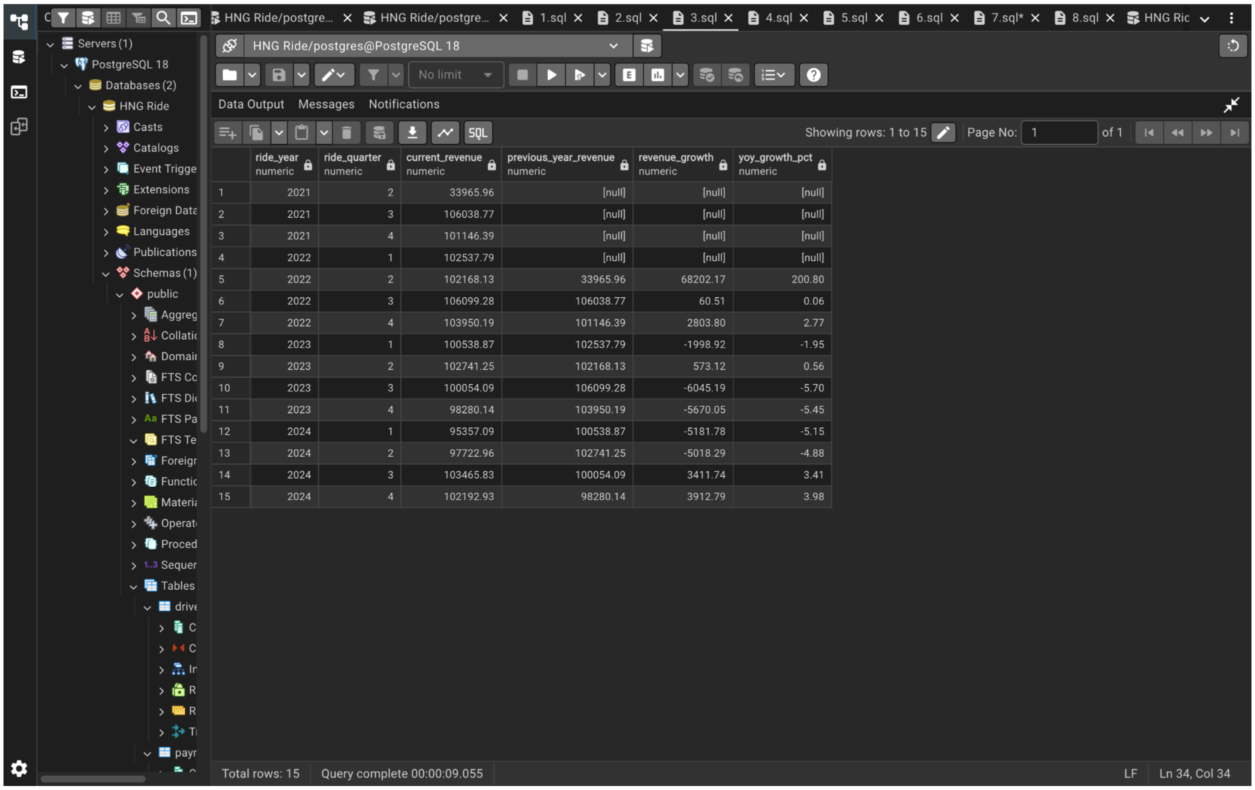
Insight: The strong retention of 1,815 riders from the 2021 cohort demonstrates excellent customer loyalty and platform stickiness. This cohort represents core, high-value customers who should be:

1. Enrolled in VIP/loyalty programs
2. Surveyed for product feedback
3. Targeted for referral incentives
4. Used as case studies to improve retention for newer cohorts

Question 3: Quarterly Revenue & Year-over-Year Growth

Query & Results:





	ride_year numeric	ride_quarter numeric	current_revenue numeric	previous_year_revenue numeric	revenue_growth numeric	yoy_growth_pct numeric
1	2021	2	33965.96	[null]	[null]	[null]
2	2021	3	106038.77	[null]	[null]	[null]
3	2021	4	101146.39	[null]	[null]	[null]
4	2022	1	102537.79	[null]	[null]	[null]
5	2022	2	102168.13	33965.96	68202.17	200.80
6	2022	3	106099.28	106038.77	60.51	0.06
7	2022	4	103950.19	101146.39	2803.80	2.77
8	2023	1	100538.87	102537.79	-1998.92	-1.95
9	2023	2	102741.25	102168.13	573.12	0.56
10	2023	3	100054.09	106099.28	-6045.19	-5.70
11	2023	4	98280.14	103950.19	-5670.05	-5.45
12	2024	1	95357.09	100538.87	-5181.78	-5.15
13	2024	2	97722.96	102741.25	-5018.29	-4.88
14	2024	3	103465.83	100054.09	3411.74	3.41
15	2024	4	102192.93	98280.14	3912.79	3.98

Total rows: 15 Query complete 00:00:09.055

- **Q2 2022 showed exceptional YoY growth: 200.80%** compared to Q2 2021
- Q2 2021 had partial data (only June-July), making comparison skewed
- **Most significant actual growth: Q2 2022** with revenue increase of $68,202.17
- **2023 showed declining trend:** Negative growth in Q1 (-1.95%), Q3 (-5.70%), and Q4 (-5.45%)
- **2024 recovery:** Q3 (+3.41%) and Q4 (+3.98%) showed positive growth returning

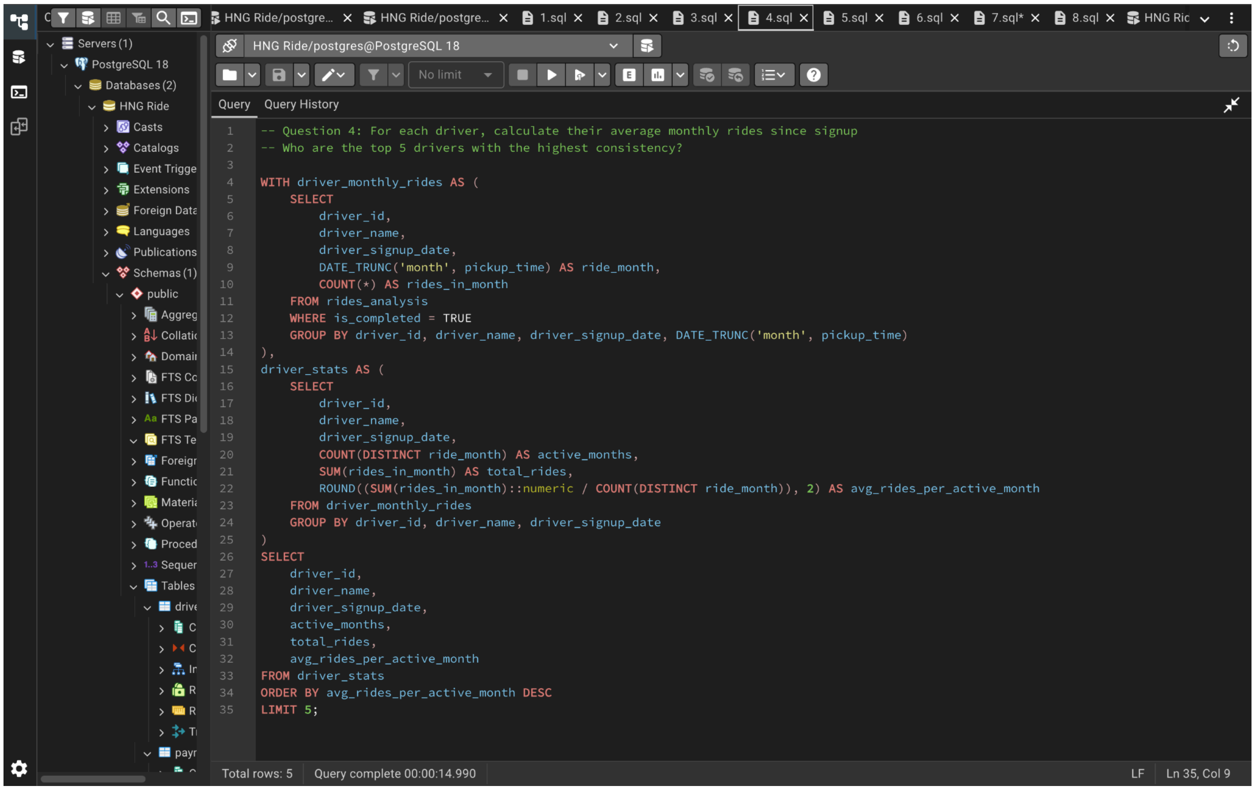
Revenue Patterns by Year:

- 2021: $241,151 (partial year from June)
- 2022: $414,755 (peak year)
- 2023: $401,614 (decline of 3.17%)
- 2024: $398,738 (slight decline of 0.72%)

Insight: CRITICAL CONCERN - The business peaked in 2022 and has been declining since. The 2023-2024 decline requires immediate investigation:

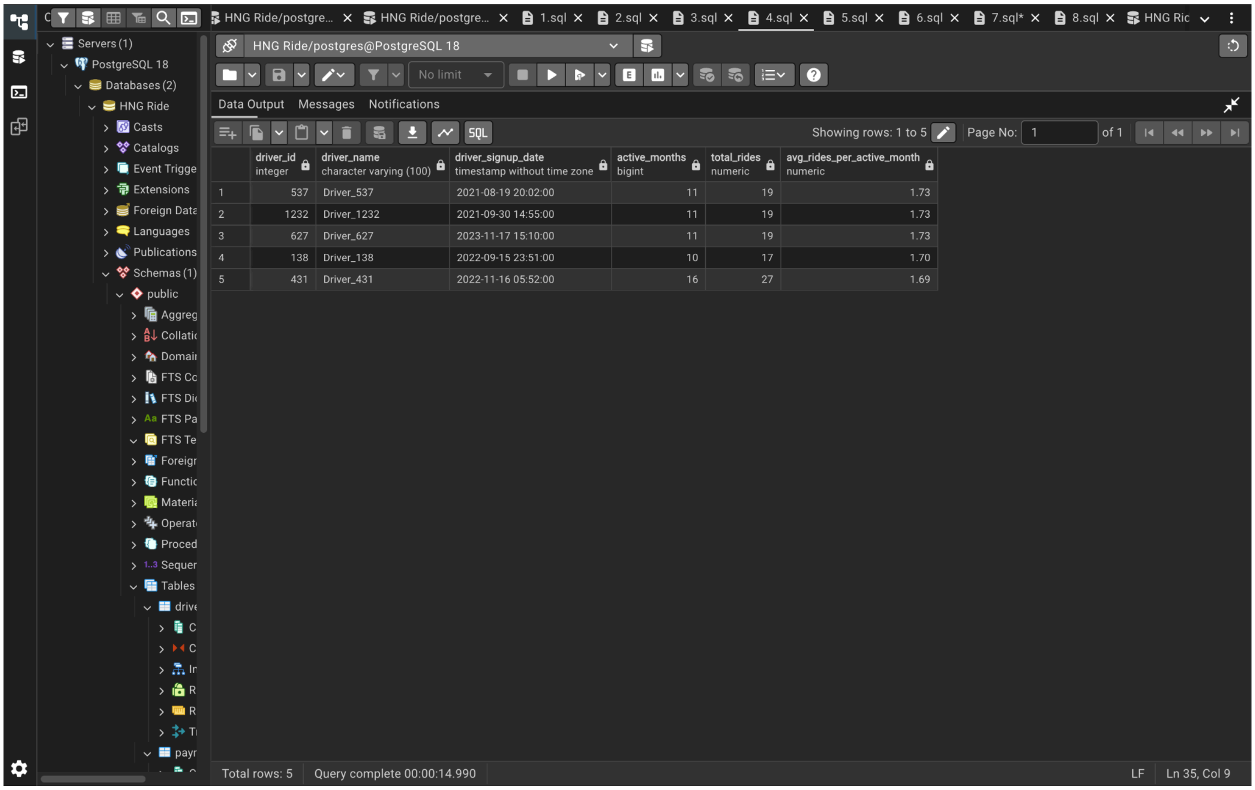
Question 4: Most Consistent Drivers

Query & Result:




```sql
-- Question 4: For each driver, calculate their average monthly rides since signup
-- Who are the top 5 drivers with the highest consistency?

WITH driver_monthly_rides AS (
    SELECT
        driver_id,
        driver_name,
        driver_signup_date,
        DATE_TRUNC('month', pickup_time) AS ride_month,
        COUNT(*) AS rides_in_month
    FROM rides_analysis
    WHERE is_completed = TRUE
    GROUP BY driver_id, driver_name, driver_signup_date, DATE_TRUNC('month', pickup_time)
),
driver_stats AS (
    SELECT
        driver_id,
        driver_name,
        driver_signup_date,
        COUNT(DISTINCT ride_month) AS active_months,
        SUM(rides_in_month) AS total_rides,
        ROUND((SUM(rides_in_month)::numeric / COUNT(DISTINCT ride_month)), 2) AS avg_rides_per_active_month
    FROM driver_monthly_rides
    GROUP BY driver_id, driver_name, driver_signup_date
)
SELECT
    driver_id,
    driver_name,
    driver_signup_date,
    active_months,
    total_rides,
    avg_rides_per_active_month
FROM driver_stats
ORDER BY avg_rides_per_active_month DESC
LIMIT 5;
```



	driver_id integer	driver_name character varying (100)	driver_signup_date timestamp without time zone	active_months bigint	total_rides numeric	avg_rides_per_active_month numeric
1	537	Driver_537	2021-08-19 20:02:00	11	19	1.73
2	1232	Driver_1232	2021-09-30 14:55:00	11	19	1.73
3	627	Driver_627	2023-11-17 15:10:00	11	19	1.73
4	138	Driver_138	2022-09-15 23:51:00	10	17	1.70
5	431	Driver_431	2022-11-16 05:52:00	16	27	1.69

Key Findings:

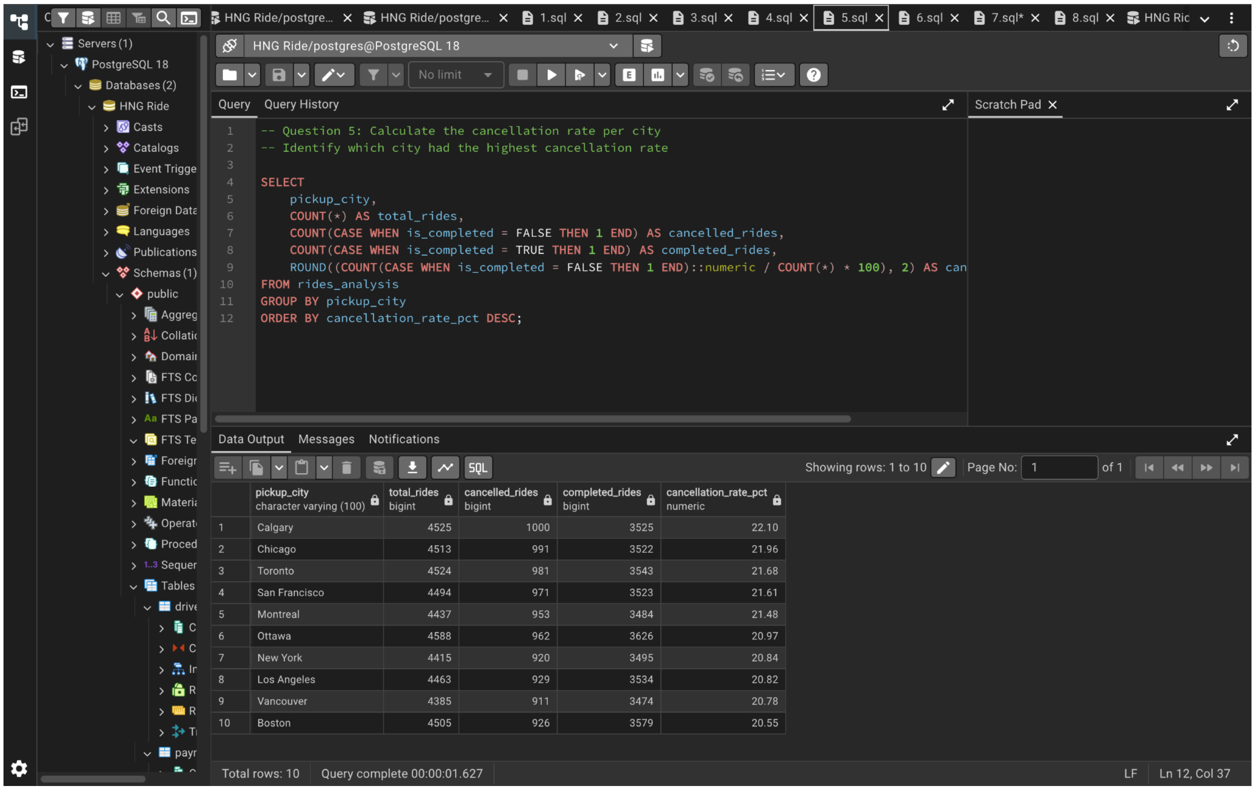
- Most consistent drivers average less than 2 rides per active month
- Driver_431 shows longest consistency (16 months)
- Low ride frequency indicates part-time driver behaviour.
- Consistency measured by sustained presence, not volume

Insight: CONCERNING FINDING - Even the "most consistent" drivers are averaging fewer than 2 rides per month. This suggests:

1. **Low driver utilisation:** Most drivers are very part-time
2. **Possible supply-demand imbalance:** Too many drivers for available rides
3. **Driver satisfaction concerns:** Low ride frequency may lead to driver churn

Question 5: Cancellation Rates by City

Query & Result:



Lowest cancellation rate: Boston at 20.55%

Key Findings:

- **ALL cities show alarming 20-22% cancellation rates**
- Calgary has the highest cancellation rate at 22.10%
- Difference between highest and lowest is only 1.55 percentage points
- Cancellation issue is systemic, not city-specific

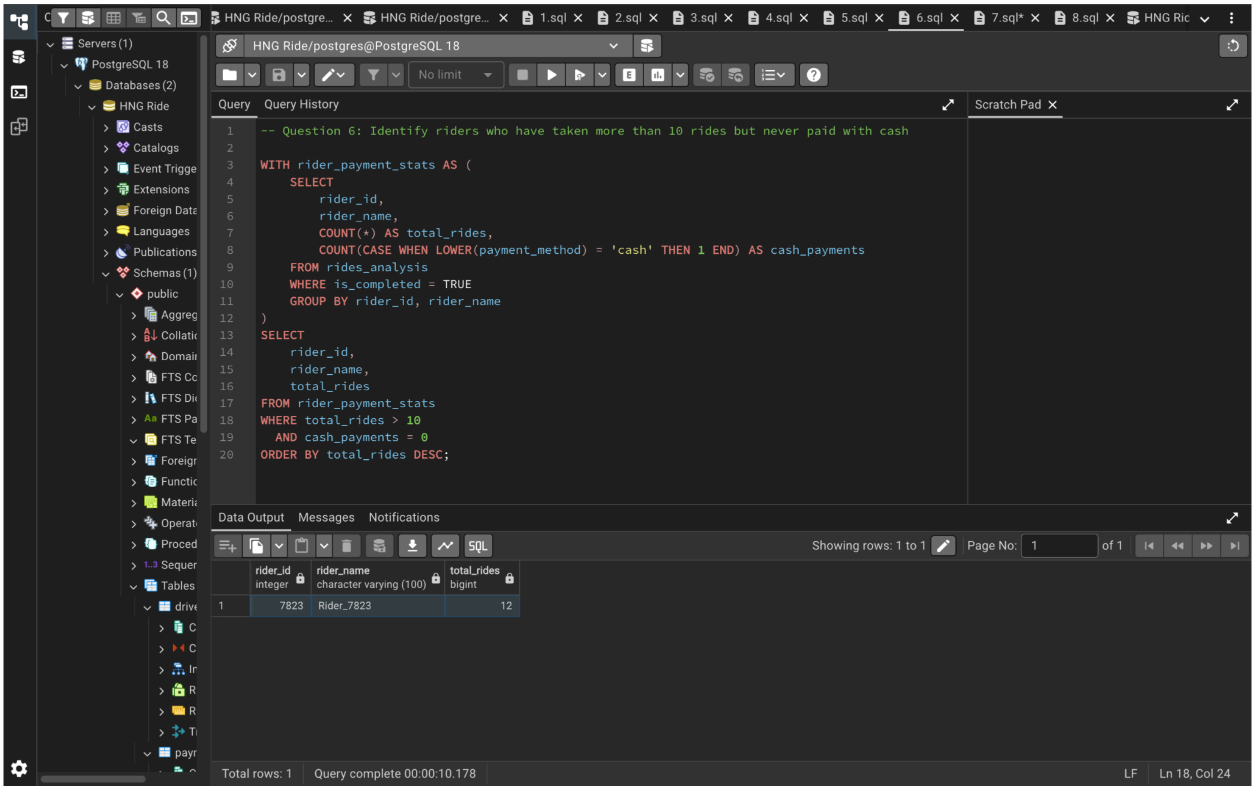
Insight: CRITICAL ISSUE - The 20-22% cancellation rate across all cities indicates a fundamental operational problem affecting the entire platform. This is not a geographic issue but a systemic one.

Impact:

- Revenue loss: ~$89,000+ in lost fares (21% of potential revenue)
- Poor customer experience
- Driver frustration and potential churn
- Reduced platform efficiency

Question 6: High-Volume Riders (No Cash Payments)

Query:



Results:

- **Only 1 rider** (Rider_7823) with 12 rides who never used cash

Key Findings:

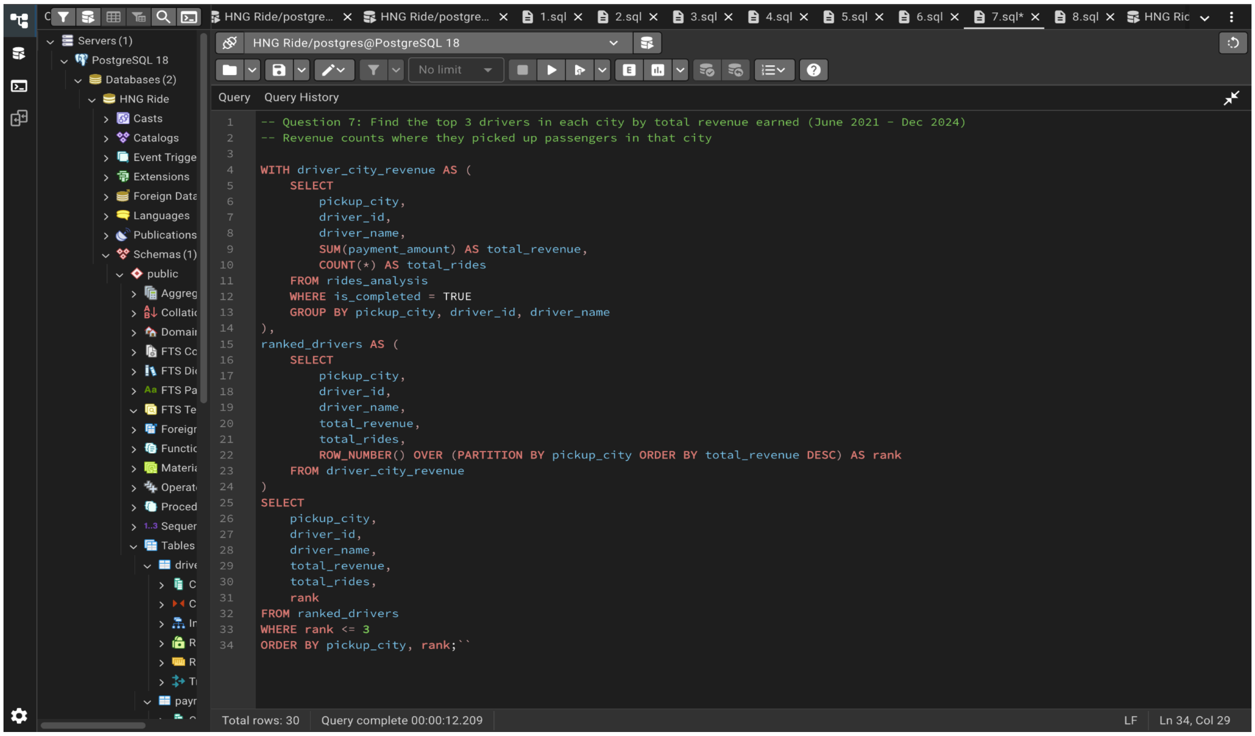
- Extremely low number of qualifying riders
- Cash remains the dominant or commonly used payment method
- Digital-only preference is rare in this user base

Insight: This finding reveals important payment behaviour patterns:

Analysis:

- The 10+ ride threshold with zero cash usage is extremely rare
- Most riders, even frequent ones, use cash at least occasionally
- Suggests either:
    1. Cash is preferred/necessary in operating regions
    2. Digital payment options may have limitations
    3. Mixed payment method usage is the norm

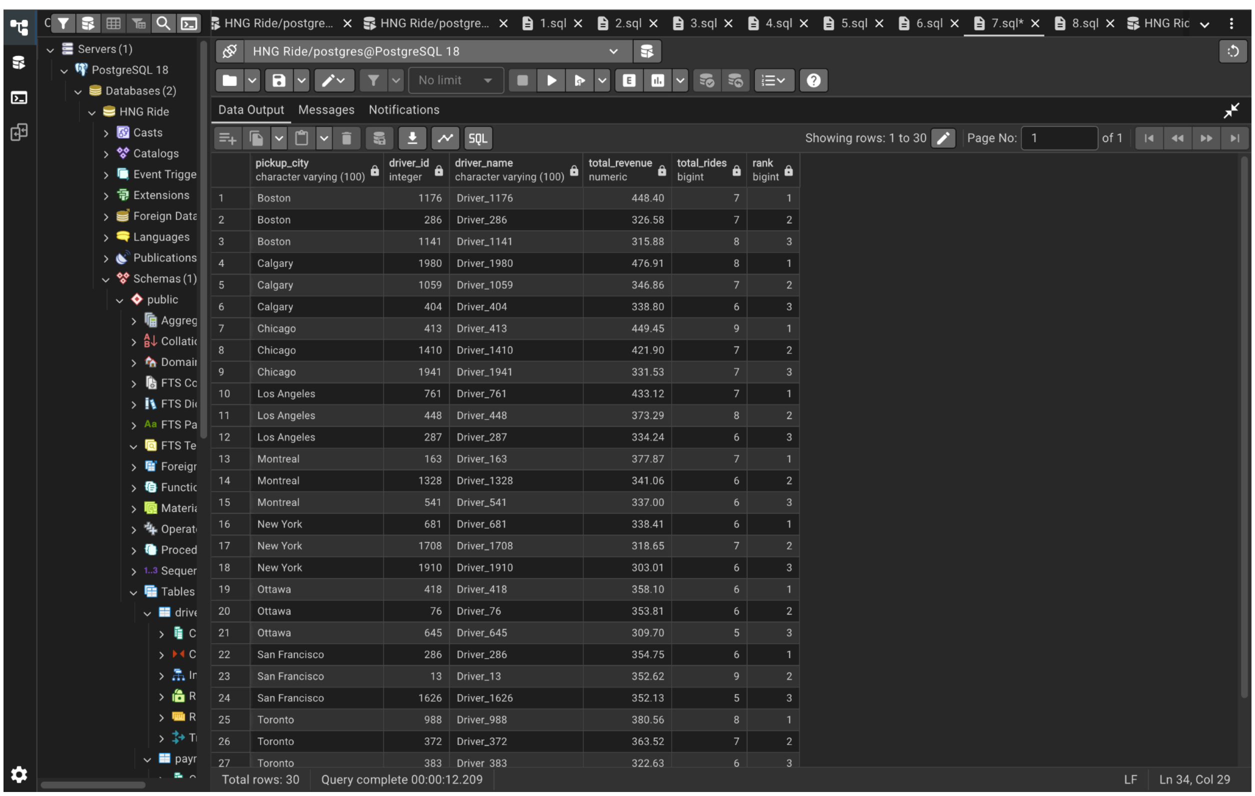
Question 7: Top Revenue Drivers by City:



```sql
-- Question 7: Find the top 3 drivers in each city by total revenue earned (June 2021 - Dec 2024)
-- Revenue counts where they picked up passengers in that city

WITH driver_city_revenue AS (
    SELECT
        pickup_city,
        driver_id,
        driver_name,
        SUM(payment_amount) AS total_revenue,
        COUNT(*) AS total_rides
    FROM rides_analysis
    WHERE is_completed = TRUE
    GROUP BY pickup_city, driver_id, driver_name
),
ranked_drivers AS (
    SELECT
        pickup_city,
        driver_id,
        driver_name,
        total_revenue,
        total_rides,
        ROW_NUMBER() OVER (PARTITION BY pickup_city ORDER BY total_revenue DESC) AS rank
    FROM driver_city_revenue
)
SELECT
    pickup_city,
    driver_id,
    driver_name,
    total_revenue,
    total_rides,
    rank
FROM ranked_drivers
WHERE rank <= 3
ORDER BY pickup_city, rank;``
```

Total rows: 30 Query complete 00:00:12.209



	pickup_city character varying (100)	driver_id integer	driver_name character varying (100)	total_revenue numeric	total_rides bigint	rank bigint
1	Boston	1176	Driver_1176	448.40	7	1
2	Boston	286	Driver_286	326.58	7	2
3	Boston	1141	Driver_1141	315.88	8	3
4	Calgary	1980	Driver_1980	476.91	8	1
5	Calgary	1059	Driver_1059	346.86	7	2
6	Calgary	404	Driver_404	338.80	6	3
7	Chicago	413	Driver_413	449.45	9	1
8	Chicago	1410	Driver_1410	421.90	7	2
9	Chicago	1941	Driver_1941	331.53	7	3
10	Los Angeles	761	Driver_761	433.12	7	1
11	Los Angeles	448	Driver_448	373.29	8	2
12	Los Angeles	287	Driver_287	334.24	6	3
13	Montreal	163	Driver_163	377.87	7	1
14	Montreal	1328	Driver_1328	341.06	6	2
15	Montreal	541	Driver_541	337.00	6	3
16	New York	681	Driver_681	338.41	6	1
17	New York	1708	Driver_1708	318.65	7	2
18	New York	1910	Driver_1910	303.01	6	3
19	Ottawa	418	Driver_418	358.10	6	1
20	Ottawa	76	Driver_76	353.81	6	2
21	Ottawa	645	Driver_645	309.70	5	3
22	San Francisco	286	Driver_286	354.75	6	1
23	San Francisco	13	Driver_13	352.62	9	2
24	San Francisco	1626	Driver_1626	352.13	5	3
25	Toronto	988	Driver_988	380.56	8	1
26	Toronto	372	Driver_372	363.52	7	2
27	Toronto	383	Driver_383	322.63	6	3

Total rows: 30 Query complete 00:00:12.209

Key Findings:

- Top performers earn $300-$477 in revenue per city
- Ride counts for top performers: 5-9 rides
- Revenue-per-ride varies significantly: $40-$80 average
- Some drivers appear in multiple cities (e.g., Driver_286 in Boston and San Francisco)

Insight:

Revenue Distribution Analysis:

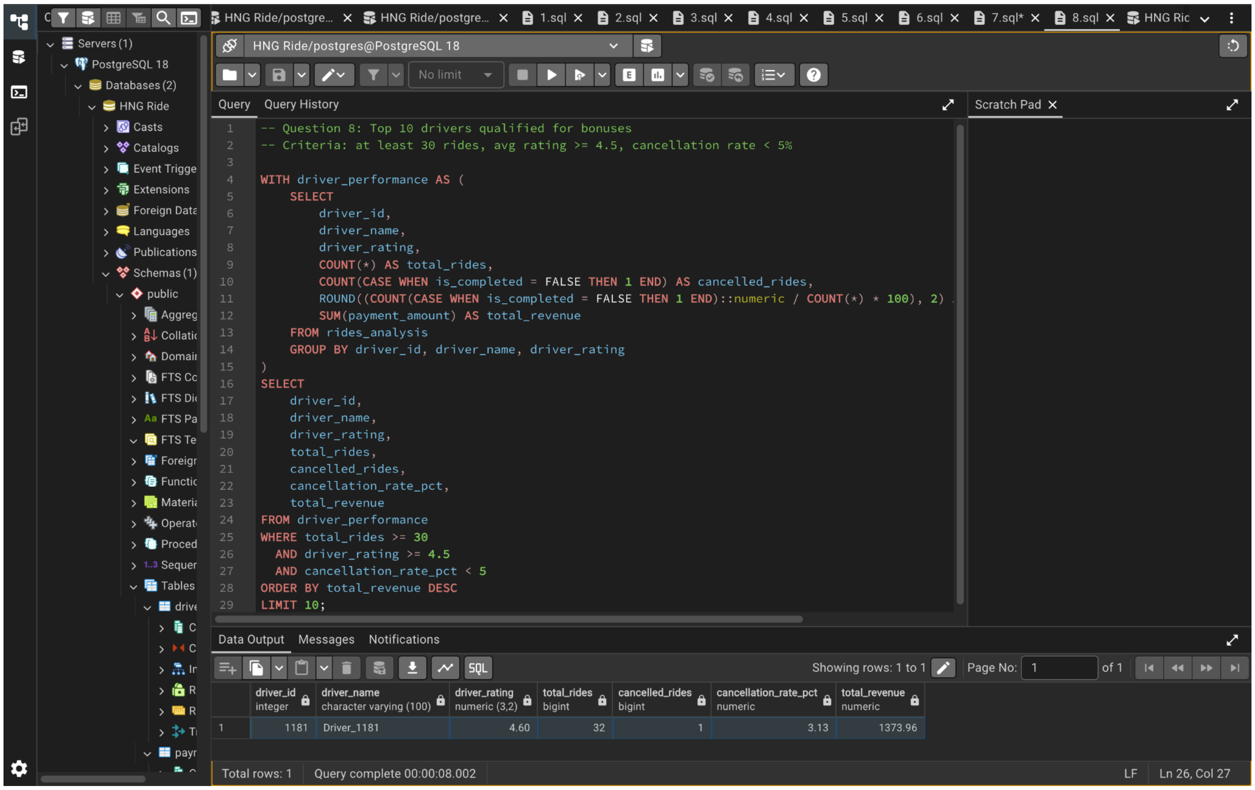
- Even top performers have very low ride volumes (5-9 rides)
- High revenue-per-ride suggests either:
    1. Long-distance premium rides
    2. Effective surge pricing capture
    3. High-value routes

Cross-City Drivers: Driver_286 appears as top performer in both Boston (#2 with $326.58) and San Francisco (#1 with $354.75), indicating:

- Multi-city operations or driver mobility
- Possible data quality issue (same driver ID in different cities)
- Opportunity for driver flexibility programs

Question 8: Bonus-Qualified Drivers

Query:



Results:

- **Only 1 driver qualifies** for the bonus: Driver_1181
  - Total rides: 32
  - Driver rating: 4.60
  - Cancellation rate: 3.13%
  - Total revenue: $1,373.96

Detailed Criteria Analysis:

From earlier investigation:

- **151 drivers** have 30+ rides (7.5% of total drivers)
- **1,123 drivers** have rating ≥ 4.5 (56% of drivers)
- **Only 45 drivers** have cancellation rate < 5% (2.25% of drivers)
- **77 drivers** meet rides + rating criteria
- **1 driver** meets rides + cancellation criteria
- **1 driver** meets all three criteria

Next Closest Competitors:

- Driver_1731: 32 rides, 4.40 rating, 6.25% cancellation (fails rating by 0.1, cancellation by 1.25%)
- Multiple 5.0 rated drivers fail due to 8-13% cancellation rates

Key Findings:

- **The <5% cancellation threshold is extremely restrictive**
- Many excellent drivers with perfect 5.0 ratings don't qualify
- The bonus program effectively disqualifies 99.95% of drivers

Insight: URGENT POLICY REVISION NEEDED

The current bonus structure is counterproductive and potentially damaging to driver morale and retention.

Problems with Current Structure:

1. **Unrealistic expectations:** Only 0.05% qualify despite 77 high-performing drivers available
2. **Demotivating:** Drivers with excellent ratings can't achieve <5% cancellation (often not their fault)
3. **Misaligned incentives:** Cancellation rate is heavily influenced by passenger behaviour and system issues
4. **Missed opportunity:** 76 other excellent drivers go unrecognised.

Impact of Current Policy:

- Loss of driver motivation
- Competitive disadvantage vs other platforms
- Unfair penalisation for passenger-initiated cancellations
- Potential driver churn

CONCLUSION

HNG Ride faces critical challenges that require immediate attention:

Most Urgent Issues:

1. **Revenue decline since 2022:** Business has contracted 3-4% and continues downward trend
2. **Systemic 21% cancellation rate:** Affecting all cities, causing significant revenue loss
3. **Overly restrictive bonus program:** Demotivating 99.95% of drivers
4. **Low driver utilisation:** Even top consistent drivers average <2 rides/month

Positive Findings:

1. **Exceptional customer retention:** 1,815 riders from 2021 cohort still active in 2024
2. **Stable operations:** 78.7% completion rate when rides aren't cancelled
3. **Geographic diversity:** Balanced operations across 10 major cities

Path Forward:

The data reveals a business at a crossroads. While customer loyalty is strong, operational inefficiencies (high cancellations), declining revenue, and driver underutilisation threaten long-term viability. The recommended actions focus on:

- **Stabilising revenue** through market repositioning
- **Improving operational efficiency** by addressing cancellations
- **Enhancing driver satisfaction** with realistic incentive programs
- **Optimising resource utilisation** through better supply-demand management

APPENDIX

Technical Notes

- **Database:** PostgreSQL 18.0
- **Analysis Tool:** pgAdmin
- **Data Quality:** Raw datasets cleaned and validated before analysis
- **Date Range:** June 1, 2021 - December 31, 2024
- **Analysis Window Note:** 2021 data begins in June (Q2), creating partial year comparison challenges

Data Sources

- drivers_raw: 2,000 records
- riders_raw: 10,000 records
- rides_raw: 50,000 records
- payments_raw: 50,000 records